10.9 SCIENCE *issues*

REPRODUCTIVE TECHNOLOGY

Making babies is not always easy and simple. Not everyone can make their own babies.

The term **infertility** describes the inability to conceive or carry a pregnancy to a live birth. About 20 per cent of all couples are infertile. One of the commonest causes of infertility is the inability of either the male or the female to produce gametes. Such a person is **sterile**.

Some of the other reasons why couples may not be able to have children are listed in the table below. Reproductive technologies have been developed to help people overcome some of these problems.

Some of the reasons why couples may not be able to have children

Type of problem	Definition/reason
gametes	Sperm or ova are not produced in sufficient quantity or quality.
impotence	Some men cannot maintain an erection during sexual intercourse.
blockage or damage	Some women may have blockages in their reproductive system (e.g. Fallopian tubes), preventing fertilisation.
miscarriage	The zygote or embryo is not maintained until the full term of the pregnancy.

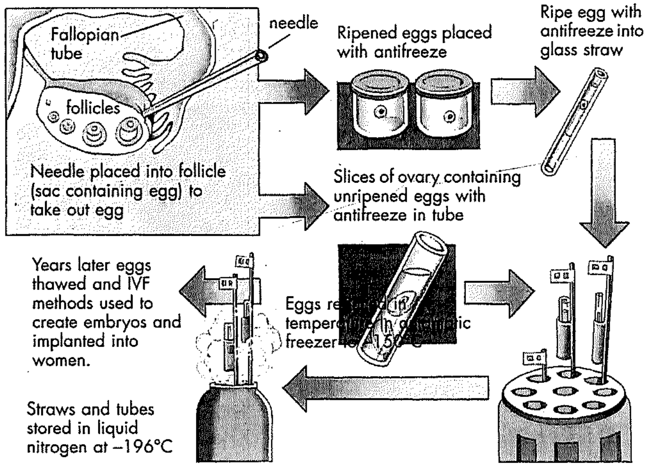
Artificial *insemination (AI)*

This technique involves injection of sperm into the woman's uterus close to the time of ovulation. The sperm may be collected from her partner, or from another male if her partner is sterile. Artificial insemination has also been used in agriculture in the production of prime farm animals, and in the breeding programs for endangered species.

In-vitro *fertilisation* (IVF)

In this technology, eggs are surgically removed using a needle, laparoscope and forceps. A laparoscope has the lens of a microscope. To improve the chances of success, the woman is often treated with drugs that cause super-ovulation. This results in several ova maturing at the same time (instead of one, as is usually the case). It is possible to freeze any fertilised eggs that are not used so that they may be available at a future time. This could enable 'twins' to be born years apart. Some women may use **donor eggs** (ova from other women).

The removed egg is then fertilised outside the mother's body in a small glass tube or dish. The sperm used has been treated so that its outer protein coat has been removed (an event that usually occurs in the female's reproductive tract). The fertilised egg is incubated in the laboratory until it is at the two- or four-cell stage. A four-cell embryo is obtained at about 35–46 hours after fertilisation.



Frozen egg technology

The embryo is then placed into the woman's uterus for implantation. Babies born using this technique are often referred to as being **test-tube babies**. In 1980, Australia's first IVF baby, a girl, was born in Melbourne; the first frozen embryo baby was also born in Melbourne, in 1984.

Some women are unable to maintain the growing embryo inside their uterus. A woman may, for example, be born without a uterus. The development of IVF technology has also opened up the field of **surrogacy**. In this situation, eggs are surgically removed from one woman and fertilised using IVF techniques. After this they are placed into another woman who undergoes the pregnancy. The woman who carries the child through to birth is called a **surrogate**.

Testing *the unborn* child

Ultrasound involves using sound waves to produce images of an unborn child inside the mother's body. During **amniocentesis**, a fine needle is inserted into the amniotic sac of the fetus and a small amount of fluid is drawn out be tested. Both techniques enable abnormalities, and also the sex of the child, to be identified.

Activities

Remember

1. Which techniques would help a couple reproduce if:
   (a) the male was infertile
   (b) the male was impotent
   (c) the female had blocked Fallopian tubes
   (d) the female had a history of miscarriages?
2. What is the difference between:
   (a) artificial insemination and in-vitro fertilisation
   (b) ultrasound and amniocentesis?
3. Outline, in point form, the steps involved in IVF.
4. What are test-tube babies? Is this an adequate name for them? Explain.

Think

(a) In groups of four or more, discuss each issue statement in the table below.
(b) Write a list of people's 'gut reactions' or immediate responses to each statement.
(c) Make a list of arguments for, and a list of arguments against, each statement.
(d) Suggest what factors influenced your opinions on these issues.
(e) Did the opinions differ between members of your group? Suggest reasons why.
(f) Report your findings back to the class, or organise a debate.
(g) Make a summary paragraph about the class's overall response for each statement.

Technique	Issue statement	Your opinion (Explain your response, with arguments for and against the issue.)
AI	• Sperm should only be used from males with a high IQ, blue eyes and red hair. • All women should be artificially inseminated with sperm selected by their parents.	
IVF	• The IVF program is too expensive and should be abandoned. • IVF technology should be used to build a superior race.	
donor gametes/ surrogacy	• Donors and surrogates should be anonymous and have no rights over the offspring produced. • Sperm should be collected from all males at the age of 18 and only this is to be used for fathering children.	
frozen embryos	• These embryos should be available to other couples if they are not used within six months. • These embryos should be developed so that they provide a supply of blood and organs for transplants.	
ultrasound/ amniocentesis	• These tests should be made compulsory to all women. Any abnormalities should result in immediate removal of the fetus. • These techniques should be used to select the gender of the child.	

10.10

Eggs on ice?

One egg or two? Imagine having your sex cells frozen, or your future child waiting patiently in a tube stored in liquid nitrogen at −196°C while you complete secondary school and settle into a career.

Reproductive technologies are breaking new boundaries and requiring new legislation. The following articles, published in a newspaper in the same week, show some of the changes that are occurring in our society.

BIRTH OF A LEGAL DILEMMA

By TERRY BROWN and HELEN CARTER

IN 1981, three tiny embryos were pinpoints of hope in a life of tragedy.

Their mother, Elsa Rios, had been shattered when her only child, Claudia, was found shot dead on her 10th birthday.

Melbourne's world-leading IVF technology offered her a second chance at motherhood, but the medical procedure only gave birth to a legal, moral and financial tangle.

Two embryos frozen after the third was unsuccessfully implanted would be the subject of litigation and legislation.

In 1983, Mrs Rios and her second husband, pilot Mario Rios, died in a light plane crash in Chile without making arrangements for the frozen embryos.

The couple were believed to have gone to Chile to adopt a child.

It was months before authorities in Melbourne knew of the couple's death, and it left a web of problems.

The Rios died without a will, leaving the embryos as potential heirs to a $1 million estate.

Mr Rios' son from a previous marriage, Mr Michael Rios, sought guardianship of the embryos and a ruling from a US court on the fate of the family fortune.

The Californian Superior Court denied a request for a guardian to be appointed for the embryos and ruled they could not be heirs to the estate.

It had been feared that another couple might try to adopt the embryos to claim the inheritance.

IVF pioneer Prof. Carl Wood said neither Michael nor Elsa's mother wanted the embryos.

"The Rios case was one of the saddest I had ever heard about," Prof. Wood said in 1990. "This couple had been through so much heartbreak, it was difficult to turn them down."

Mario's low sperm count meant donor sperm was needed but Elsa's eggs could be used.

Three embryos resulted. Two were frozen and left at the Queen Victoria Medical Centre. The other was implanted in Elsa's womb but the pregnancy only lasted six days.

Prof. Wood said yesterday the couple had returned overseas, leaving their embryos here, because they were trying to adopt.

"I remember them very well. They were a lovely couple, very enthusiastic and religious. They were wealthy but a very understanding couple and encouraging to the program," he said.

"They were very generous and I'm sure they would have donated their embryos to another couple if they'd known, but they can't be offered because they weren't HIV tested."

Scientists said even if the embryos were viable, there was a less than 5 per cent chance of surviving thawing because at the time they were frozen the freezing technique was still being developed.

Prof. Wood said past laws did not set a time limit.

Ms Howlett said under the old legislation Monash IVF could not dispose of the Rios embryos.

The new legislation says:

AN embryo cannot be kept in storage unless it will be used for an IVF procedure and can only be stored for five years, although applications for extension can be made.

AN embryo from a dead person cannot be implanted into a woman.

EMBRYOS can be removed from storage only to be used in treatment, research or if one or both parents are dead.

Victorian Health Department spokesman Graeme Walker said Monash IVF did not have to apply for permission to let them die.

"Under the legislation, once they've been there five years, they are free to discard them," he said.

Prof. Wood said embryos removed from the nitrogen tank disintegrated in a few hours.

He said it was unlikely the Rios embryos could grow into a child.

"When the Rioses came to Melbourne, no IVF babies had been born from the technology, although pregnancies had been achieved," he said.

"The embryos could not be donated to another couple because it was before HIV testing of parents, so we could say the embryos are free of contamination".

Ms Howlett said since the case couples must sign forms detailing what they wanted to happen to embryos if they died or divorced.

Prof. Wood would ask Monash IVF's board whether it would allow the embryos to "succumb" and if a request would be put to the ITA and Victorian Health Minister Marie Tehan. He said in the past embryos were allowed to die if a couple had enough children from other embryos.

But he said the Rios embryos' status was uncertain.

ITA chairman Prof. Louis Waller said in 1984 the Rios embryos should be allowed to die with dignity.

Former Supreme Court judge and Royal Commissioner Ken Marks, chairman of the Standing Review and Advisory Committee on Infertility, said as a general rule embryos could not be disposed of.

He said the new law seemed to imply you could dispose of embryos by taking them out of storage.

St Vincent's Hospital ethicist Dr Bernard Clarke said if embryos were still viable they had the potential to be a human being.

"I believe it is not proper to dispose of embryos," he said.

Source: *Herald Sun*, 10 February 1996

HOPE AT FIRST HUMAN EGG BANK

By HELEN CARTER, medical reporter

WOMEN with cancer can now have their own children using the world's first human egg bank.

The bank, officially opened yesterday at the Royal Women's Hospital in Melbourne, uses a method perfected by its scientists to freeze human eggs.

Women, adolescents, children and babies with cancer and other fertility-threatening diseases can have eggs removed before treatment or their condition destroys them.

Ripened eggs or an ovary or ovary section containing unripened eggs are frozen and years later thawed for use in standard IVF.

Human embryos can be frozen and stored but people must be married to do so. Now people can store eggs years before they meet a partner.

Reproductive biology unit scientist Debra Gook, who developed the method, said 10 women aged 18 to mid-30s had frozen eggs in storage.

Nine had cancer and one the fertility-affecting condition, endometriosis.

The bank has had inquiries from women wanting to store eggs of their young daughters with cancer and Turner's Syndrome, where ovaries shut down before 10.

Unit director Dr Andrew Speirs said some people regained fertility after cancer treatment but others did not.

He said sometimes women did not have optimal treatment for fear it would interfere with fertility but now they could have the best and not worry.

Dr Speirs said Victorian law allowed frozen eggs and sperm to be stored for 10 years and after that an application could extend time.

Radiographer Rose Ford, 33, daily gave mammograms to women checking for breast cancer and in February 1994 found a lump and was diagnosed herself.

"The bank is so exciting because this is for once a door opening instead of shutting," she said.

Ms Ford plans to freeze her eggs if they are viable.

FOR egg and sperm freezing inquiries, call 9344–2371.

Source: *Herald Sun*, 6 February 1996

World first: reproductive scientist Debra Gook and cancer sufferer Rose Ford.

Activities

Think

1. What is a human egg bank?
2. (a) Who may and who may not use the frozen egg bank?
   (b) Do you agree with this? Give reasons for your opinions.
3. Suggest the advantages and disadvantages of the frozen egg bank.
4. If you were given the opportunity next week, would you have your sperm or eggs frozen? Give reasons for your response.
5. How many Rios embryos were frozen?
6. When and why were the Rios embryos frozen?
7. Which type of infertility made it difficult for the Rioses to conceive naturally?
8. Make a list of the problems or dilemmas which occurred after Mr and Mrs Rios died in the plane crash in 1983.
9. Suggest the types of laws or regulations that you would impose on frozen embryo technology.
10. Summarise the new IVF legislation laws stated in the article.
11. Do you agree or disagree with each of these laws? Give a reason for your opinions.
12. Once out of the nitrogen tank, how long would the embryos stay alive for?
13. State two reasons why the Rios embryos would be unlikely to grow into a child.
14. What do you think should happen to the Rios embryos? Why?

2

Quotes from children/young people conceived using ART

Note: DI refers to donor insemination

"I think if the person has reached 18 years and the parents have still not told them, that they are an adult and they have a right to know they are donor-conceived and that is their true identity."

(Narelle, donor-conceived adult; told by her parents at 15)

"Can you imagine having a blood test in your adulthood only to discover that your blood group doesn't match that of either of your parents? Or even worse, discovering the secret from family friends, or in the middle of a heated argument?"

(Lauren, donor-conceived adult, told by her parents at 9)

"I wish my parents' attitude had been healthy enough that they could have spared me the anguish of having it sprung on me by a third party. I am sure that the 'secret' would have surfaced eventually, because it turns out that most of my extended family had known about it from the very start. I cannot adequately describe how it feels to discover that everyone except me had known. I realise that they were protecting me (and my mother and dad and perhaps themselves) and that intentions generally were all good."

(Melody, donor-conceived adult, told by her godmother at 33)

"How to tell? I haven't got the vaguest idea. I think if my mother had not just couched it so much in fear. If she had just told us straight away, perhaps taken us to a nice park and worked up to it. Instead, she said she was going to tell us something and that it was really big and that she was scared about it. She kind of built up to it for a few days. We thought it was something really bad, so when she finally did tell us we thought it wasn't so bad. I guess it depends a lot on the family and upon the kids."

(Marta, donor-conceived adult, told by her mother at 15)

"My mother did not tell me of my origins until I was 37 years old (in 1983), a few days after the death of my younger DI brother. … I was confused, angry, relieved, hurt and embarrassed and yet full of sympathy for my parents. … I had long suspected that my dad was not my genetic father. … I was most surprised that my conception was DI not from adultery, as I had assumed as a teenager. So my long-time shame about my mother vanished but was replaced by anger that she had not trusted me with the truth."

(William, donor-conceived adult)

"The only lingering emotion I have from my mother's disclosure thirteen years ago is regret. … I regret that my parents did not have the courage or insight of DI parents today who choose to share the truth with their children. I regret the lost chance to have lived a life in openness. … It would have been wonderful."

(William, donor-conceived adult, told by his mother at 37)

"I think that everyone should be able to have contact with their biological parents."

(Narelle, donor-conceived adult; told by her parents at 15)

"When I was 18 I attempted to track down information about my biological father. The information which I feel is important to me includes: medical history, racial origins and physical characteristics; whether he and his parents are still alive; and information concerning half-siblings born through donated genetic material and through other relationships. In this category I seek information about: the number of half-siblings, their age, gender, and whereabouts. Yes, my list does include updated information. For me the link with my donor does not just stop at the time of my conception. Information about the donor's entire life should be consistently updated until the offspring wish to access the information, and even after that time."

(Lauren, donor-conceived adult, told by her parents when she was 9)

"I think it is preposterous that anyone would expect me not to wonder about and want to know who this man is. … This does NOT mean that I desire a personal relationship with the donor or his family members. … I do not imagine or wish for a 'replacement' father. … My curiosity is mainly genealogical in nature: a 'family tree' project, if you will."

(Melody, donor-conceived adult, told by her godmother at 33)

"My parents told my brother and I about our conception when I was 9 and he was 12. Our parents sat us down and said, 'We have something important to tell you.' To this my brother's immediate response was, 'I'm adopted, aren't I?' This illustrates the fact that, when secrets are kept, the children often grow up sensing that something is different within their family. The funny thing is that this is not necessarily due to what their parents do say, but as a result of what the parents don't say. For example, they never say, 'You've got your father's eyes and your grandmother's personality'. … In my family we are very comfortable with the situation of my conception. In fact it is often the subject of humour and jokes. My brother and I often use the donor as a scapegoat and the source of our less attractive traits! I'm quite sure I must have inherited his nose because I didn't get Mum's small one."

(Lauren, donor-conceived adult, told by her parents at 9)

Taken from *Telling it your way: A guide for parents of donor-conceived adolescents*
Dr Maggie Kirkman Professor Doreen Rosenthal Louise Johnson Published by the Infertility Treatment Authority Victoria, Australia

3

Common terms and their meanings

ART: Assisted reproductive treatment. Technological and other methods to achieve pregnancy.

Azoospermia: A condition in men where no sperm are found in the semen.

Chromosome: The gene packaging in a cell. Every cell has two copies of each chromosome – one from each parent.

Clinical pregnancy: A pregnancy confirmed by a blood test and ultrasound, usually performed at around six to eight weeks.

Cytoplasm: All the material within a cell other than the nucleus.

DNA: Deoxyribonucleic acid; molecule that contains the genetic code that determines the characteristics of the organism.

DNA fragmentation: Strands of DNA broken into smaller pieces.

Embryo: The stages of development of a fertilised egg up to 11 weeks gestation.

Fertilisation: The moment when the DNA from a sperm and egg combine.

Gamete: A mature reproductive cell, such as a sperm or egg (ovum).

Gene: Section of DNA that contains the instructions (genetic code) to make molecules and proteins. Every person has two copies of each gene – one from each parent.

Genetic abnormality: Missing or additional portions of DNA in the chromosomes within a cell, or an abnormal number of chromosomes.

Hypospadias: A malformation in boys where the urinary opening is not located at the head of the penis, but instead is found along the underside of the penis or, in rare cases, within the scrotum.

In-vitro: Literally means "in glass". Refers to techniques being performed in laboratory containers such as petri dishes, test tubes and flasks.

In-vitro fertilisation (IVF): An assisted reproductive treatment where an egg and sperm are combined in the laboratory before being transferred to the woman's uterus.

Male factor infertility: Infertility related to sperm count and/or motility and/or morphology, or ability to engage in intercourse or ejaculate.

Morphology: Shape and structure.

Motility: Ability to move.

Nucleus: The compartment inside a cell that contains the genetic material (DNA)

Oocyte: An immature egg cell, which matures into an ovum.

Oligospermia: Low sperm count (less than 15 million sperm/ml).

Ovum (plural ova): A mature egg. The female reproductive cell.

Semen: The fluid that is produced in the male reproductive tract that carries the sperm.

Spermatozoon (plural spermatozoa): Sperm, the male reproductive cell.

Testicular sperm aspiration (TESA): A method where sperm are collected directly from the testicles under anaesthetic.

Testicular sperm extraction (TESE): A method where a biopsy is collected from the testicles under anaesthetic to collect sperm.

Vas deferens: Tube that carries sperm from the testicles to the urethra for ejaculation.

Zona pellucida: The outer layer of the egg.

Zygote: The single cell formed by the combination of DNA from a sperm and an egg through fertilisation.

Artificial Reproductive Technologies (ART) notes

in vitro fertilisation – embryo transfer (IVF - ET) ('classic' IVF)

Success rate:

Method:

Pros:

Cons:

Other moral/legal issues:

gamete intrafallopian transfer (GIFT)

Success rate:

Method:

Pros:

Cons:

Other moral/legal issues:

zygote intrafallopian transfer (ZIFT)

Success rate:

Method:

Pros:

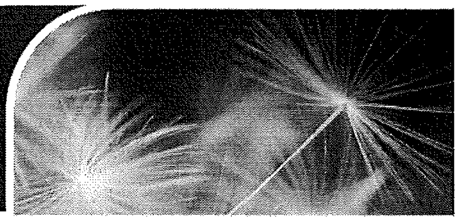
Cons:

Other moral/legal issues:

Other Types of IVF

frozen embryo transfer (IVF - FET)	Intra-cytoplasmic sperm injection (IVF-ICSI)
Success rate:	Success rate:
Method (how it differs from IVF-ET):	Method (how it differs from IVF-ET):
Pros:	Pros:
Cons:	Cons:
Other moral/legal issues:	Other moral/legal issues:

What is assisted reproductive treatment (ART)?



VARTA provides independent information about fertility treatment. This brochure describes the main options available. When considering fertility tests and treatments, VARTA recommends asking the following questions:

- Do I really need this test, treatment or procedure?
- What are the risks?
- Are there simpler, safer options?
- What happens if I don't do anything?
- What are the costs?
- Will it improve my chance of having a baby?

What is ART?

Assisted reproductive treatment (ART) is used to help people achieve pregnancy. ART can be used:

- as infertility treatment for heterosexual couples
- to help people who identify as LGBTIQ+, and single people have children
- by people who can't become pregnant or carry a pregnancy without treatment
- to reduce the risk of a child inheriting a genetic disease or abnormality.

Types of assisted reproductive treatment

Simple treatment options

Ovulation induction

Ovulation induction (OI) can be used if somebody is not ovulating or not ovulating regularly. It involves taking a hormone medication (tablets or injections) to stimulate ovulation. The response to the hormones is monitored with ultrasound and when the time is right, an injection is given to trigger ovulation (the release of the egg). Timing intercourse to coincide with ovulation offers the chance of pregnancy.

Artificial insemination

Artificial insemination (AI), which is sometimes called intrauterine insemination (IUI), involves insertion of sperm into a person's uterus at or just before the time of ovulation. AI can help couples with so called unexplained infertility or when there is a minor sperm abnormality.

AI can be performed during a natural menstrual cycle, or in combination with ovulation induction if a person has irregular menstrual cycles. If a pregnancy is not achieved after a few AI attempts, IVF or ICSI may be needed.

What is assisted reproductive treatment (ART)?

Advanced treatment options

In-vitro fertilisation (IVF)

During IVF, hormone injections are used to stimulate the ovaries to produce multiple eggs. When the eggs are mature, they are retrieved in an ultrasound-guided procedure under light anaesthetic. The eggs and sperm are placed together in a laboratory to hopefully form embryos. Three to five days later, if an embryo has formed, it can be either placed in the uterus (fresh embryo transfer) or frozen to be transferred into the uterus at a later time (frozen embryo transfer). Any additional embryos can be frozen and used later if the first transfer does not work.

Hormone stimulation	Fertility drugs are given to develop several eggs (stimulated cycle).
Egg retrieval	Eggs are collected.
Embryo development	Sperm is added to the eggs for embryos to form. Sometimes more than one embryo develops that is suitable for transfer. When there are several embryos available for transfer, one is usually transferred* and the remainder frozen for later use if there is no pregnancy. Sometimes all embryos are frozen.
Embryo transfer	An embryo is placed in the uterus. Two weeks later a blood test is done to see if the embryo has implanted and the woman is pregnant.
Clinical pregnancy	If the test is positive an ultrasound is done two to three weeks later to verify the pregnancy. A pregnancy at this early stage does not guarantee the birth of a baby because some pregnancies end in miscarriage.

* Single embryo transfer (transferring one embryo at a time) is considered the gold standard of practice in IVF to minimise the risk of multiple pregnancy which is associated with higher risks for both mother and babies.

Intracytoplasmic sperm injection (ICSI)

Intracytoplasmic sperm injection (ICSI) is used to overcome sperm problems. ICSI follows the same process as IVF, except ICSI involves the direct injection of a single sperm into each egg to hopefully achieve fertilisation (whereas in IVF the egg and sperm are left in a petri dish to fertilise on their own).

Because it requires technically advanced equipment, there are additional costs for ICSI. For couples with male factor infertility, ICSI is needed to fertilise the eggs and give them a chance of having a baby. But for couples who don't have male factor infertility, the chance of a baby is no better with ICSI than with IVF.

Using donor sperm, eggs, or embryos

There are many reasons why donor sperm, eggs or embryos may be needed. Your fertility specialist and the clinic counsellor will provide you with information about using donor sperm, eggs or embryos and can discuss the social, emotional, and legal considerations.

What is assisted reproductive treatment (ART)?

Donor sperm
Donor insemination (DI) may be used by:

- single people
- same-sex couples
- couples where the male partner does not produce any sperm or no normal sperm
- couples with a high risk of a man passing on a genetic disease or abnormality to a child.

The process of DI is the same as artificial insemination (AI). If the female partner also has an infertility problem, donor sperm can be used in IVF treatment.

Donor embryos
Donor embryos can be used if a person or couple requires both donor sperm and donor eggs to achieve a pregnancy. In rare cases, some people donate embryos they don't need to other people. The recipient takes hormones in preparation for the embryo transfer and when the lining in the uterus is ready, embryos are thawed and transferred.

Donor eggs
Treatment with donor eggs may be needed when:

- a person doesn't produce eggs, or their eggs are of low quality. This may be due to age or premature menopause (ovarian failure)
- a person has experienced several miscarriages, or
- there is a high risk of a person passing on a genetic disease or abnormality to a child.

In these cases, the egg donor has hormone injections to produce several eggs. When the eggs are mature, they are retrieved and sperm from the recipient's partner or a donor is added to the eggs. Two to five days later, when embryos have formed, one is inserted into the recipient's uterus. In the two to three weeks leading up to the embryo transfer, the recipient takes hormones to make sure the lining in the uterus is ready for an embryo to implant. If a pregnancy is confirmed, the hormone treatment continues for another eight to 10 weeks.

Preimplantation genetic testing (PGT)
When preimplantation genetic testing (PGT) is used, embryos are generated through the process of IVF or ICSI and then a few cells are removed from the embryo and tested for normality.

For people who are at high risk of passing on a genetic condition to their offspring, embryos can be tested before they are transferred to make sure only unaffected embryos are transferred. PGT for monogenic/single gene defects (PGT-M) is used to identify embryos that are not affected by a 'faulty' gene that can lead to disease. PGT for chromosomal structural rearrangements (PGT-SR) is used to identify embryos that have an incorrect amount of genetic material. PGT-M and PGT-SR are also known as preimplantation genetic diagnosis (PGD).

Some clinics offer preimplantation genetic testing for aneuploidy (PGT-A). PGT-A, is an add-on used to help choose embryos with the right number of chromosomes. While PGT-A can reduce the time it takes to get pregnant by eliminating embryos unlikely to produce a live birth, it doesn't increase the overall chance of having a baby. It's also costly and there is a small risk that healthy embryos are discarded.

Surrogacy
Surrogacy involves a person carrying a child for another person or couple with the intention of giving the child to that person or couple after birth. VARTA has resources to assist people considering surrogacy.

Pre-implantation genetic testing explained

Pre-implantation genetic testing (PGT) is a technique used in IVF to help people reduce their risk of passing on a known genetic condition.
There are two types of PGT:

- **PGT for monogenic/single gene defects (PGT-M)** is used to identify embryos that are not affected by a 'faulty' gene that can lead to disease.
- **PGT for chromosomal structural rearrangements (PGT-SR)** is used to identify embryos that have the correct amount of genetic material.

PGT-M and PGT-SR are also known as **pre-implantation genetic diagnosis (PGD).**

How is PGT done?

Hormone stimulation	Fertility drugs are given to develop a number of eggs (stimulated cycle).
Egg retrieval	Eggs are retrieved from the ovaries and sperm are added to the eggs to create embryos.
Embryo development and biopsy	Embryos are grown in the laboratory for 5-6 days. At this time, two types of cells are distinguishable: the cells that will develop into the placenta and the cells that will become the baby. A few cells are removed from the future placenta for genetic testing to minimise the risk that the developing fetus is harmed.
PGT	Genetic testing is performed on the cells that are removed. The time it takes to get PGT results can vary from days to weeks and depends on the method used and where the testing takes place. Embryos are frozen until test results are available.
Embryo transfer	If the test shows that there is one or more normal embryos, one is thawed and transferred to the woman's uterus. Any remaining normal embryos will be kept frozen for transfer later if the first transfer does not lead to a pregnancy.

VARTA
Victorian Assisted Reproductive Treatment Authority

Pre-implantation genetic testing explained

Who needs which test?

PGT for monogenic conditions (PGT-M)

Some individuals carry a 'faulty' gene that may not affect them but can cause severe genetic conditions in their offspring (e.g. cystic fibrosis or Huntington's disease). They may become aware of this when they have a child affected by the condition or if they have a family member who is affected.

A PGT-M test can be designed for almost any genetic disorder caused by a single 'faulty' gene, as long as the location of the gene causing the disorder is known.

PGT for structural rearrangements (PGT-SR)

Some individuals have chromosomal rearrangements that do not affect their health but can affect their chance of having a healthy baby. They may become aware of this as a result of chromosomal testing, which is often recommended after repeated miscarriages or the birth of a child with a chromosomal abnormality.

PGT-SR testing can be used for any chromosome rearrangement that a parent is known to have and helps identify embryos with the correct amount and arrangement of chromosomal material.

Advantages and disadvantages of PGT-M and PGT-SR

It is important to know that **PGT does not guarantee the birth of a healthy baby.** Women who become pregnant are advised to undergo prenatal testing (e.g. DNA testing or chorionic villus sampling) to confirm PGT results.

Here are some of the advantages and disadvantages of embryo testing. For more detailed information and to find out if PGT is for you, please speak to your doctor or genetic counsellor.

Advantages

- For people who are known to carry a faulty gene or a chromosomal rearrangement, it reduces the risk of having a child with a serious health condition.
- For women who have had previous miscarriages due to a known structural rearrangement in their chromosomes, PGT-SR reduces the risk of miscarriage.
- It reduces the risk of having to make difficult decisions later, including whether to terminate or continue a pregnancy with a fetus affected by a genetic or chromosomal condition.[1]

Disadvantages

- PGT requires IVF treatment, which fertile people do not otherwise need.
- The cost of PGT is not covered by Medicare and is in addition to the costs of IVF.
- PGT only tests for specific conditions, so it does not eliminate the risk of a child having another condition that the test does not cover.
- Embryos may not survive the biopsy procedure.[2]
- Due to technical challenges, there is a small chance that the test results may not reflect the true health of the embryo.[3]
- In the case of chromosomal rearrangements, the embryo may contain a mixture of cells with normal and abnormal chromosome arrangements. This is called **mosaicism.** This can cause a false positive or false negative PGT result.[4]
  - **A false positive result** means that the few cells that are tested show abnormalities, while the remaining cells may be normal. Based on PGT alone, an embryo that may have led to the birth of a healthy baby may be discarded.
  - **A false negative result** means that the few cells that are tested are normal while the remaining cells may have abnormalities. Based on the test result, a chromosomally abnormal embryo may be transferred.
- Embryos may not survive the thawing process.
- Sometimes none of the embryos are suitable for transfer due to a genetic abnormality. Your doctor will discuss your results and options with you.

References

1. Lamb, B., Johnson, E., Francis, L., Fagan, M., Riches, N., Wilson, A., Johnstone, E. (2018). ***Pre-implantation genetic testing: decisional factors to accept or decline among in vitro fertilization patients.*** Journal of Assisted Reproduction and Genetics, 35(9), 1605-1612.
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3. Brezina, P. R., & Kutteh, W. H. (2015). ***Clinical applications of preimplantation genetic testing.*** BMJ, 350, g7611.
4. Victor, A.R., Tyndall, J.C., Brake, A.J., Lepkowsky, L.T., et al. (2019). ***One hundred mosaic embryos transferred prospectively in a single clinic: exploring when and why they result in healthy pregnancies.*** Fertility and Sterility, 111(2):280-93.

Costs of IVF

The services that clinics include in their 'global' fee per cycle vary, so if you know what services and medications will be required in your treatment, you will be in a better position to ask clinics what they would include in a cycle fee and what you would have to pay for separately.

Before starting IVF treatment you will have:

Medical tests

After initial tests to find out why you have not been able to get pregnant, you may need to have further medical tests before treatment can begin.

These might include:

- screening blood tests (blood group; German measles; chicken pox; hepatitis B and C; HIV and syphilis)
- other blood tests (depending on the cause of infertility, you may need to have some hormone levels checked, such as Anti-Mullerian Hormone (AMH), which gives an indication of a woman's ovarian reserve).
- semen analysis
- ultrasound examinations.

Medicare rebates are available for the majority of these assessments, so make sure you have a current referral from your doctor (GP). Check with your clinic if there are additional costs for any pre-treatment tests or counselling.

Counselling

In the Australian state of Victoria it is a legal requirement for people undergoing IVF to see an infertility counsellor before starting treatment.

Information session(s)

At most clinics, you will have an information session with a nurse who will make sure you know what an IVF cycle involves, that you understand what medications you will be taking and what you need to do as part of your IVF treatment. You may also meet with administrative staff who explain the financial aspects of your treatment.

Steps in the IVF cycle

There are a number of stages involved in IVF treatment. It will help to be familiar with the stages of treatment to understand the costs involved for each stage.

- **Step 1:** some treatments involve taking medications that 'switch off' the woman's natural cycle of egg production in the ovaries (down-regulation).
- **Step 2:** stimulation of the ovaries to produce more than one egg (ovulation induction).
- **Step 3:** collecting the eggs from the woman's ovaries (egg pick-up).
- **Step 4:** collecting sperm from the male partner (if applicable).
- **Step 5:** fertilising the eggs with a male partner's or donor's sperm in a laboratory. This will entail either the standard IVF or the Intracytoplasmic sperm injection (ICSI) process. ICSI entails further costs.
- **Step 6:** incubating the fertilised eggs for a few days. Fertilised eggs that have started to develop are called embryos.
- **Step 7:** after a few days the embryo(s) is/are put into the woman's uterus (embryo transfer).
- **Step 8:** if there are additional suitable embryos these are frozen and can be used in subsequent cycles.
- **Step 9:** approximately two weeks after embryo transfer a pregnancy test will determine if the treatment has been successful.

Costs of IVF

IVF cycle

During the treatment cycle the IVF cycle fee covers the following:

- Specialist and nursing consultations
- Counselling while on treatment
- Ultrasound scans
- Blood tests
- Embryology/laboratory services
- Semen preparation
- Egg collection (a private health fund may contribute to this cost)
- Embryo transfer (a private health fund may contribute to this cost)
- Some treatment medications (Medicare covers the cost of some treatment medications)
- Pregnancy test

The following services may incur additional costs:

- Pre-treatment counselling
- Pre-treatment tests
- ICSI (intracytoplasmic sperm injection)
- Extended embryo culture to blastocyst stage (if required)
- Excess embryo freezing
- Ongoing embryo storage
- Day surgery or hospital costs (for egg pick up and embryo transfer) – anaesthetist and bed fees
- Specialist consultation fees
- Early pregnancy care

Frozen embryo transfer cycle

If the IVF cycle results in more than one or two embryos assessed as suitable for transfer, those remaining can be frozen for later use. In a frozen embryo transfer cycle, the embryos are thawed and transferred when the lining in the uterus has the best chance for embryo implantation.

This is either a couple of days after ovulation in a natural cycle or after a couple of weeks of taking estrogen tablets to build up the lining in the uterus.

During the frozen embryo transfer cycle the IVF fee covers the following:

Blood tests
Ultrasounds
Embryology/laboratory services

You should check with the clinic whether the fee includes:

Specialist and nurse consultations
Medications

Intrauterine insemination cycle

Treatment involves inserting the male's concentrated semen through the neck of the womb and into the uterus close to the time of ovulation.

During the treatment cycle the Medicare rebate should include the following:

Blood tests
Ultrasound scans
Some treatment medications
Sperm preparation
Insemination

Additional costs:

ICSI (Intracytoplasmic sperm injection)

ICSI is used when there is male-related infertility such as a low sperm count. ICSI follows the same process as IVF, except ICSI involves the direct injection of a single sperm into each egg to achieve fertilisation. There is a Medicare rebate for ICSI but most clinics charge more than the rebate so there is also an out-of-pocket cost.

Other laboratory procedures

In addition to the standard embryology procedures, your fertility specialist might recommend other procedures such as preimplantation genetic diagnosis (PGD). If additional procedures are recommended ask the clinic about the costs associated with these.

Hospital/ Day surgery

With most clinics, patients' egg collection takes place in a private hospital or day surgery centre. Medicare does not provide a rebate for private hospital fees but if you have private health insurance and have served the mandatory waiting period your health fund may cover a portion of the hospital costs. Contact your health fund directly to ask.

Anaesthetist

You may also incur a fee for the anaesthetist attending to you during the egg collection procedure. If you have private health insurance and have served the mandatory waiting period, you might be eligible for a rebate from your health fund.

Medication

Your fertility specialist might prescribe medication that is not covered by the Pharmaceutical Benefit Scheme (PBS). Ask the clinic about fees associated with all medications that the fertility specialist prescribes.

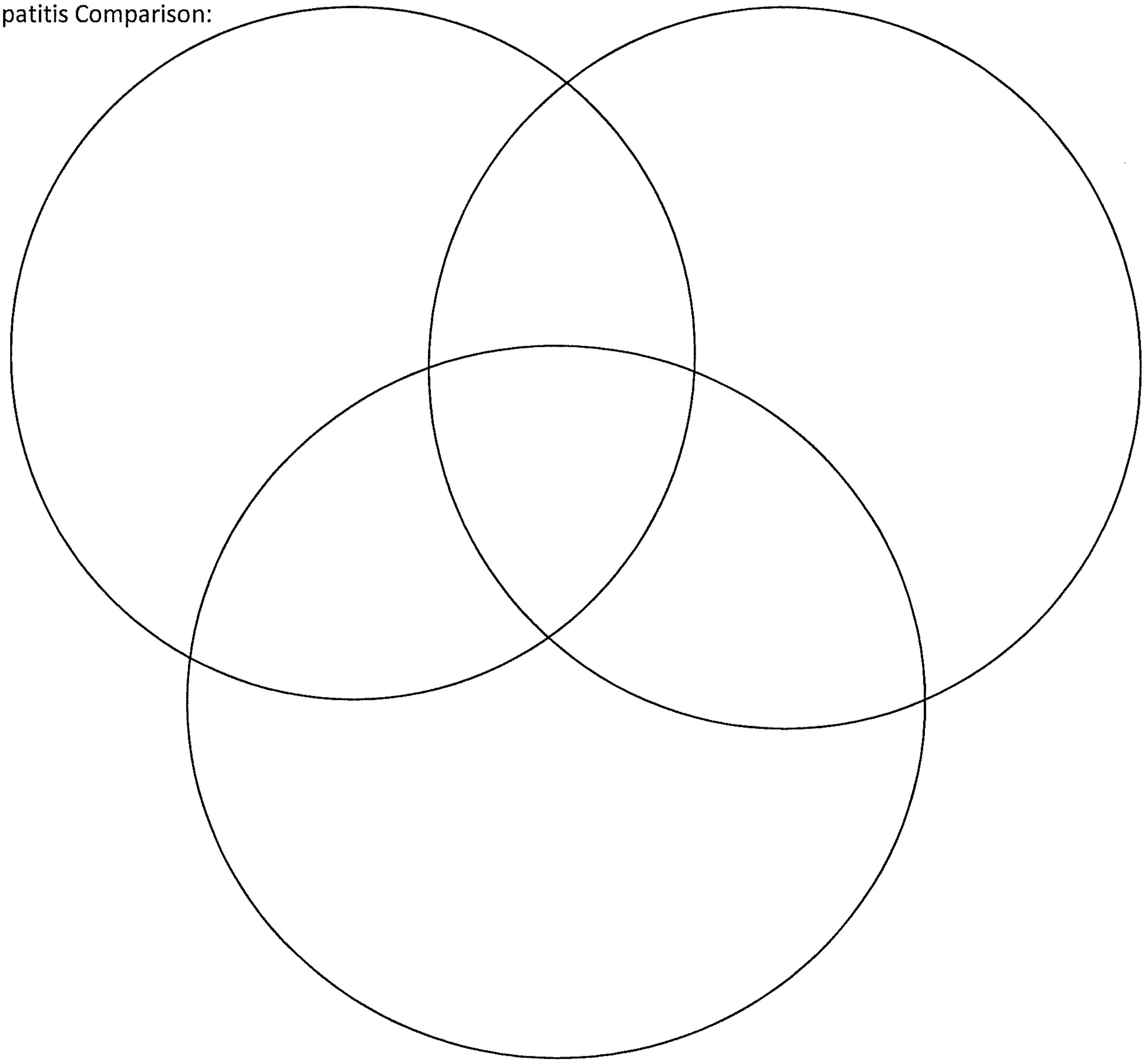
Donor eggs/sperm/embryos

If you need to use donor eggs, sperm or embryos, this will incur additional costs.

Examples of STI's

Virus	Bacteria	Yeast	Lice or Mites	Other

Hepatitis Comparison:



WORKSHEET 1

Activity 2: How Much Do You Know?

Sexual Health Questionnaire

Please answer 'yes' or 'no' by putting a tick in the Yes or No column

	Yes	No
a) Could a person get HIV (the AIDS virus) by sharing a needle and syringe with someone when injecting drugs?	....	....
b) Could a woman get HIV (the AIDS virus) through having sex with a man?	....	....
c) If someone with HIV coughs or sneezes near other people, could they get the virus?	....	....
d) Could a man get HIV through having sex with a man?	....	....
e) Could a person get HIV from mosquitoes?	....	....
f) If a woman with HIV is pregnant, could her baby become infected with HIV?	....	....
g) Could a person get HIV by hugging someone who has it?	....	....
h) Does the pill (birth control) protect a woman from HIV infection?	....	....
i) Could a man get HIV through having sex with a woman?	....	....
j) If condoms are used during sex does this help to protect people from getting HIV?	....	....
k) Could someone who looks healthy pass on HIV infection?	....	....

Please answer the remaining questions by putting a tick in the True or False column

	True	False
1. A man can have a sexually transmitted infection without any obvious symptoms.	....	....
2. A woman can have a sexually transmitted infection without any obvious symptoms.	....	....
3. Apart from HIV, all sexually transmitted infections can be cured.	....	....

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Activity 2: How Much Do You Know?

Please answer the remaining questions by putting a tick in the True or False column

	True	False
4. Chlamydia is a sexually transmitted infection that affects only women.	....	....
5. Chlamydia can lead to sterility among women.	....	....
6. Once a person has caught genital herpes, then they will always have the virus.	....	....
7. People who always use condoms are safe from all STIs	....	....
8. Gonorrhoea can be transmitted during oral sex.	....	....
9. Genital warts can only be spread by sexual intercourse.	....	....
10. HIV only infects gay men and injecting drug users.	....	....
11. Cold sores and genital herpes can be caused by the same virus.	....	....
12. Hepatitis C has no long term effects on your health.	....	....
13. It is possible to be vaccinated against Hepatitis A.	....	....
14. It is possible to be vaccinated against Hepatitis B.	....	....
15. It is possible to be vaccinated against Hepatitis C.	....	....
16. People who have injected drugs are not at risk for Hepatitis C.	....	....
17. Hepatitis C can be transmitted by tattooing and body piercing.	....	....
18. Hepatitis B can be transmitted sexually.	....	....
19. All people that have Hepatitis C. can be cured.	....	....
20. Hepatitis C. can be transmitted by sharing razors and toothbrushes.	....	....

Smith, A., Dyson, S., Agius, P., Mitchell, A., & Pitts, M. (2003) *Secondary Students and Sexual Health 2002*. Melbourne: Australian Research Centre in Sex, Health and Society, La Trobe University.

WORKSHEET 1

Activity 2: How Much Do You Know?

Australian Data from the Third National Survey of Australian Year 10 School Students, HIV/AIDS and Sexual Health.

Yes or No questions	% of students who answered questions correctly Male	Female
a) Could a person get HIV (the AIDS virus) by sharing a needle and syringe with someone when injecting drugs?	96.6	96.1
b) Could a woman get HIV (the AIDS virus) through having sex with a man?	94.3	96.0
c) If someone with HIV coughs or sneezes near other people, could they get the virus?	82.2	86.7
d) Could a man get HIV through having sex with a man?	85.7	83.1
e) Could a person get HIV from mosquitoes?	38.6	42.8
f) If a woman with HIV is pregnant, could her baby become infected with HIV?	67.9	73.7
g) Could a person get HIV by hugging someone who has it?	95.6	98.6
h) Does the pill (birth control) protect a woman from HIV infection?	85.4	89.8
i) Could a man get HIV through having sex with a woman?	86.6	90.3
j) If condoms are used during sex does this help to protect people from getting HIV?	88.6	86.0
k) Could someone who looks healthy pass on HIV infection?	80.3	83.4
True or False questions		
1. A man can have a sexually transmitted infection without any obvious symptoms.	69.9	82.7
2. A woman can have a sexually transmitted infection without any obvious symptoms.	69.7	83.9
3. Apart from HIV, all sexually transmitted infections can be cured.	52.6	61.2

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Activity 2: How Much Do You Know?

True or False questions	% of students who answered questions correctly Male	Female
4. Chlamydia is a sexually transmitted infection that affects only women.	13.7	15.6
5. Chlamydia can lead to sterility among women.	23.7	37.8
6. Once a person has caught genital herpes, then they will always have the virus.	41.6	53.5
7. People who always use condoms are safe from all STIs	69.4	74.4
8. Gonorrhoea can be transmitted during oral sex.	34.2	41.7
9. Genital warts can only be spread by sexual intercourse.	29.0	45.2
10. HIV only infects gay men and injecting drug users.	73.1	85.4
11. Cold sores and genital herpes can be caused by the same virus.	35.2	51.3
12. Hepatitis C has no long term effects on your health.	49.7	54.3
13. It is possible to be vaccinated against Hepatitis A.	46.9	57.4
14. It is possible to be vaccinated against Hepatitis B.	59.8	74.4
15. It is possible to be vaccinated against Hepatitis C.	10.1	11.1
16. People who have injected drugs are not at risk for Hepatitis C.	63.9	70.7
17. Hepatitis C can be transmitted by tattooing and body piercing.	42.8	52.7
18. Hepatitis B can be transmitted sexually.	45.8	35.8
19. All people that have Hepatitis C. can be cured.	35.8	44.1
20. Hepatitis C. can be transmitted by sharing razors and toothbrushes...	34.2	29.1

Smith, A., Dyson, S., Agius, P., Mitchell, A., & Pitts, M. (2003) *Secondary Students and Sexual Health 2002*. Melbourne: Australian Research Centre in Sex, Health and Society, La Trobe University.

WORKSHEET 2

Activity 9: Who Can I Trust?

Where Do I Get My Information?

1. Put a tick next to the sources you have used or found have provided you with most of your information and ideas about sex, sexuality and relationships.

2. Circle the sources that you trust.

Mum	☐	Pamphlets/posters	☐	Radio	☐
Dad	☐	Health education at school	☐	Teachers	☐
Sister	☐	Television	☐	Doctor	☐
Brother	☐	Female friends	☐	Boyfriend/girlfriend	☐
Books/magazines	☐	Male friends	☐	School nurses	☐

3. If you wanted some information about STIs, which of the following sources would you use? Why?

Mum	☐	Pamphlets/posters	☐	Radio	☐
Dad	☐	Health education at school	☐	Teachers	☐
Sister	☐	Television	☐	Doctor	☐
Brother	☐	Female friends	☐	Boyfriend/girlfriend	☐
Books/magazines	☐	Male friends	☐	School nurses	☐

WORKSHEET 13

WORKSHEET 13

Activity 9: Who Can I Trust?

4. If you thought you had a STI, which of the following sources would you go to for advice? Why?

Mum	☐	Pamphlets/posters	☐	Radio	☐
Dad	☐	Health education at school	☐	Teachers	☐
Sister	☐	Television	☐	Doctor	☐
Brother	☐	Female friends	☐	Boyfriend/girlfriend	☐
Books/magazines	☐	Male friends	☐	School nurses	☐

5. If you wanted to talk about relationship issues such as, love, attraction, starting or ending a relationship and so on, which would you use? Why?

Mum	☐	Pamphlets/posters	☐	Radio	☐
Dad	☐	Health education at school	☐	Teachers	☐
Sister	☐	Television	☐	Doctor	☐
Brother	☐	Female friends	☐	Boyfriend/girlfriend	☐
Books/magazines	☐	Male friends	☐	School nurses	☐

6. If you wanted factual information about sex, such as pregnancy, sexual practices, wet dreams, and so on, which would you use? Why?

Mum	☐	Pamphlets/posters	☐	Radio	☐
Dad	☐	Health education at school	☐	Teachers	☐
Sister	☐	Television	☐	Doctor	☐
Brother	☐	Female friends	☐	Boyfriend/girlfriend	☐
Books/magazines	☐	Male friends	☐	School nurses	☐

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STI Basics Chart

DISEASE	TRANSMISSION	TYPES OF SEXUAL CONTACT THAT MAY PRESENT A RISK OF CONTRACTING THE DISEASE	COMMON SYMPTOMS	POSSIBLE COMPLICATIONS	TREATMENT
CHLAMYDIA (Bacteria)	Semen, pre-ejaculate (pre-cum), vaginal fluid.	Oral sex (mouth-penis, mouth-vagina) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Often no noticeable symptoms. Potential for itching, discharge or burning during urination or ejaculation, pain in the lower abdomen or back, pain during intercourse, discharge from the vagina, bleeding between menstrual periods, nausea, or fever.	If left untreated, may lead to infection of the testicles or pelvic inflammatory disease (PID) in women, a serious medical condition that can lead to infertility. May cause infertility even without symptoms. Can be transmitted from mother to newborn during childbirth.	Curable with antibiotics.
GENITAL HERPES (Virus)	Skin-to-skin contact (usually genital), saliva may transmit virus from the mouth or lips. Transmission is possible even without an outbreak of sores.	Oral sex (mouth-penis, mouth-vagina, mouth-anus) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Often no noticeable symptoms. May cause one or more sores, blisters, pimples, bumps, or a rash around mouth, genitals or anus, itching, burning, or tingling in either the genital area or the mouth, a fever, swollen glands or stiff neck and headache. May have repeated outbreaks that are generally less severe than the original.	May result in chronic painful condition particularly for people who have a weakened immune system. Can be transmitted from mother to newborn during childbirth.	No cure but medications can reduce the frequency and duration of outbreaks.
GENITAL WARTS (HPV) (Human papillomavirus)	Skin-to-skin contact (usually genital). Transmission is possible even without visible warts.	Oral sex (mouth-penis, mouth-vagina, mouth-anus) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Often no noticeable symptoms but may cause raised or flat growths around genitals or anus that are usually flesh colored or whitish in appearance.	Certain strains of HPV are considered risk factors for cervical cancer. In very rare cases, it can be transmitted from mother to newborn during childbirth.	No cure but warts can be removed using creams, surgery, cryosurgery (freezing), or laser treatment. There is now a vaccine to prevent certain types of HPV infection, including types that cause cervical cancer.
GONORRHEA (Bacteria)	Semen, pre-ejaculate (pre-cum), vaginal fluid.	Oral sex (mouth-penis, mouth-vagina) Vaginal sex (vagina-penis) Anal sex (anus-penis)	May have discharge or burning during urination or ejaculation, pain in the lower abdomen or back, pain during intercourse, discharge from the vagina, bleeding between menstrual periods, nausea, or fever. For women, there are often no noticeable symptoms.	If left untreated, may lead to infection of the testicles or pelvic inflammatory disease (PID) in women, a serious medical condition that can lead to infertility. Can be transmitted from mother to newborn during childbirth.	Curable with antibiotics.
HEPATITIS A (Virus)	Feces.	Oral sex (mouth-penis, mouth-vagina, mouth-anus)	Often no noticeable symptoms but may cause fever, tiredness, aches, loss of appetite, nausea, abdominal pain, dark urine, and jaundice (yellowing of the skin and eyeballs).	In rare cases, may lead to severe liver infection and death.	Nearly all infections resolve on their own. There are vaccines to prevent hepatitis A.

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STI Basics Chart

DISEASE	TRANSMISSION	TYPES OF SEXUAL CONTACT THAT MAY PRESENT A RISK OF CONTRACTING THE DISEASE	COMMON SYMPTOMS	POSSIBLE COMPLICATIONS	TREATMENT
HEPATITIS B (Virus)	Blood, semen, pre-ejaculate (pre-cum), vaginal fluid.	Oral sex (mouth-penis, mouth-vagina) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Often no noticeable symptoms but may cause mild flu-like symptoms including fever, tiredness, aches, loss of appetite, nausea, abdominal pain, dark urine, and jaundice (yellowing of the skin and eyeballs).	Can lead to chronic infection, cirrhosis (scarring of liver tissue) and cancer of the liver. Can be transmitted from mother to newborn during childbirth.	Nearly all infections resolve on their own but medications may be used to treat chronic illness. Alcohol and certain medicines should be avoided to prevent further liver damage. There are vaccines to prevent hepatitis B.
HIV/AIDS (Human Immuno-deficiency Virus)	Blood, semen, pre-ejaculate (pre-cum), vaginal fluid, breast milk.	Oral sex (mouth-penis, mouth-vagina) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Often no symptoms for years but may cause fever, chills and sweats, fatigue, appetite loss, weight loss, muscle and joint pain, long-lasting sore throat, swollen lymph nodes, diarrhea, yeast infections, and skin sores.	Over time, can lead to acquired immune deficiency syndrome (AIDS), which can leave a body vulnerable to other infections or cancers normally controlled by a healthy immune system. Can be transmitted from mother to newborn during childbirth.	No cure or vaccine for HIV or AIDS. There are medications that allow people to live with HIV or AIDS for longer periods of time.
SCABIES & PUBIC LICE OR CRABS (Parasite)	Skin-to-skin contact (usually prolonged sexual contact), although in rare cases, can spread by contact with clothes, towels, bedding, and other personal items that were recently in contact with an infected person	Oral sex (mouth-penis, mouth-vagina) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Intense itching usually in genital area, visible crab eggs (small, oval-shaped, beads) attached to the base of hair, may have dark or bluish spots on skin in the infested area as a result of bites.	Scratching area may lead to secondary bacterial infections.	Medicated shampoos and creams will kill mites/lice on the body. In addition, need to thoroughly clean all clothing, towels and bedding to prevent reinfection.
SYPHILIS (Bacteria)	Skin-to-skin contact (between syphilis sore and penis, vagina, anus or mouth).	Oral sex (mouth-penis, mouth-vagina) Vaginal sex (vagina-penis) Anal sex (anus-penis)	Painless sore on or around penis, vagina, mouth or anus; rash over the entire body or on the hands and soles of the feet, fever, swollen lymph glands, patchy hair loss, headaches, weight loss, muscle aches, and tiredness.	If left untreated, may damage heart, eyes, central nervous system and other organs. Can be transmitted from mother to fetus prior to birth.	Curable with antibiotics.
TRICHOMONIASIS (Parasite)	Semen, pre-ejaculate (pre-cum), vaginal fluid.	Vaginal Sex (vagina-penis, vagina-vagina)	Women may experience frothy, yellow-green vaginal discharge, discomfort during intercourse and urination, irritation and itching in the genital area and in rare cases, lower abdominal pain. Most men do not experience symptoms but may have irritation inside the penis, mild discharge, or slight burning during urination or ejaculation.	If left untreated, on rare occasions, leads to pelvic inflammatory disease (PID) in women, a serious medical condition that can lead to infertility.	Curable with antibiotics.

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